

Hao Zheng

Recent M.S. Graduate, Tsinghua University | Transportation Systems, Optimization, and Machine Learning

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CORE METHOD AREAS

Machine Learning: Time Series Forecasting Graph Neural Networks Sequence-to-Sequence Models Transfer Learning
Knowledge-Guided Learning Multi-Task Learning Discrete Choice Modeling

Optimization & Operations Research: Mixed-Integer Programming Bilevel Optimization Column Generation
Branch-and-Price VRP/VRPTW ALNS Tabu Search

Transportation & Infrastructure Modeling: Space-Time Network Modeling Traffic Assignment Microscopic Traffic Simulation
Pavement Performance Prediction Pavement Maintenance Optimization

PUBLICATIONS

- **Hao Zheng**, Han Zheng, George Giannopoulos. KNOWCAST: Domain-knowledge-integrated deep learning for metro OD forecasting and path flow estimation. DOI:10.1016/j.trc.2026.105710 🌐 *Transportation Research Part C: Emerging Technologies*, 189, 105710. 2026
Method focus: Graph Neural Networks Sequence-to-Sequence Models Knowledge-Guided Learning
- **Hao Zheng**, Han Zheng*, George Giannopoulos. Integrated Routing of Trucks and Robots for Multi-Modal Urban Express Logistics. 🌐 *Presented at the TRB 105th Annual Meeting, 2026; Transportation Research Record, Accepted for publication.* 2026
Method focus: Space-Time Network Modeling Branch-and-Price Column Generation C#/Gurobi
- Jiantao Li, Yian Zhao, **Hao Zheng**, Runhua Guo. Vehicle Vibration Modelling via Road Profile Reconstruction and Simulation: Coupled Analysis of IRI, Speed, and Ride Comfort for Driving Assistance. DOI:10.1080/14680629.2025.2574300. *Road Materials and Pavement Design*. 2025
Method focus: Road Profile Reconstruction Vehicle Dynamics Simulation IRI-Based Modeling
- **Hao Zheng**, Xingchen Zhang, Junhua Chen. Study on Customized Shuttle Transit Mode Responding to Spatiotemporal Inhomogeneous Demand in Super-Peak. *Information*, 12(10), 429. DOI:10.3390/info12100429 🌐 2021
Method focus: Pickup-and-Delivery with Time Windows ALNS Tabu Search
- **Hao Zheng**, Yun Wang*, Xuedong Yan, Yunwei Li. Dynamic Planning Model and Solution Algorithm for the Demand-responsive Customized Bus Route. *Proceedings of the World Transport Convention / World Transport Engineering and Technology Forum (WTC2021)*, People's Communications Press, 2021. ISBN: 9787114173554. 🌐
Method focus: Two-Stage Planning Dynamic Insertion Route Re-Optimization
- Yuyan Tan, Yibo Li, Ruxin Wang, Xiwei Mi, Yaxuan Li, **Hao Zheng**, Yu Ke, Yan Wang. Improving Synchronization in High-Speed Railway and Air Intermodality: Integrated Train Timetable Rescheduling and Passenger Flow Forecasting. *IEEE Transactions on Intelligent Transportation Systems*, 23(3), 2651–2667. DOI:10.1109/TITS.2021.3137410. 2022
Method focus: Timetable Rescheduling Passenger Flow Forecasting Multi-Objective Optimization

MANUSCRIPTS UNDER REVIEW AND WORKING PAPERS

- **Hao Zheng**, Jiantao Li, Yian Zhao, Xi Zhu, Runhua Guo*. An Operational Scheduling Framework for Pavement Maintenance Activities: From Rule-Based Task Generation to Branch-and-Price-Based Route Optimization. 🌐 *Revised manuscript submitted after major revision, International Journal of Pavement Engineering.* 2026
Method focus: VRPTW Road Network Modeling Branch-and-Price Column Generation
- **Hao Zheng**, Xi Zhu, Jiantao Li, Long Wen, Weining Zhou, Runhua Guo*. Public-Data-Assisted Staged Transfer Learning for Local Pavement Condition Prediction across Shared and Local Indicators. 🌐 *Submitted to International Journal of Pavement Engineering.* 2026
Method focus: Transfer Learning Multi-Task Learning Sequence-to-Sequence Models
- **Hao Zheng**, Han Zheng*, George A. Giannopoulos. DISCERN: Discrete Choice with Embedded Representation Networks. *Working paper, 2026.* 2026
Method focus: Discrete Choice Models Representation Learning Gradient-Based Estimation

SELECTED RESEARCH EXPERIENCE

- **Computational Benchmarking of Scalable Traffic Assignment on Open Transportation Networks** 2025 – Present
Research Collaborator with Prof. Xuesong Zhou

- Built an experimental pipeline converting TNTP-style benchmark networks into unified link-demand formats and generating path sets for path-based traffic assignment.
- Evaluated path-compression strategies across compression ratios and demand levels, comparing solver time, objective gaps, and link-flow fit against baseline solutions.
- **Bilevel Optimization Framework for Tolling Strategy in Autonomous Networks**  06/2025 – 10/2025
MIT-UF-NEU Summer Research Camp / Supervisor: Prof. Jinhua Zhao, MIT; Mentor: Dr. Xuan Jiang
 - Designed an end-to-end bilevel optimization workflow for tolling strategy design, modeling interactions between infrastructure operators and autonomous vehicle fleets.
 - Implemented the Python-based upper-level optimization loop and re-engineered a C++/CUDA microscopic traffic simulator to support toll-responsive agent behavior.
- **Intelligent Road Asset Management: Prediction and Decision-Support Models** 2023 – 2025
Digital Transportation Management Platform / Supervisor: Prof. Runhua Guo
 - Developed core prediction and maintenance decision modules for a digital road asset management platform, integrating pavement condition forecasting with maintenance strategy optimization.
 - Built GRU/attention-based pavement performance prediction models using climate, traffic, material, and LTPP data, and implemented budget- and performance-constrained maintenance decision models.
- **Customized Bus and Air–Rail Intermodal Optimization** 2020 – 2022
National University Student Innovation & Entrepreneurship Development Program / Project Leader
 - Led the development of a two-stage optimization framework for air–rail passenger transfers, combining high-speed rail timetable adjustment with customized bus route planning.
 - Modeled the airport–station connection as a vehicle routing problem with time windows and transfers, and developed a hybrid ALNS–Tabu Search algorithm; the resulting software received a national software copyright.
- **Additional Applied Transportation and Infrastructure Research** 2022 – 2024
 - Contributed to research on green transportation and infrastructure lifecycle management, including intelligent inspection, pavement performance prediction, and maintenance decision-making models.
 - Conducted comparative analysis of highway design and acceptance standards for international transportation infrastructure adaptation.

EDUCATION

- **Tsinghua University**, School of Civil Engineering, Master of Science, GPA: 3.86 09/2022 – 06/2025
Thesis: *A Comprehensive Full-Process Planning Model for Pavement Performance Forecasting and Maintenance Strategy Implementation under Missing Data Conditions.*
- **Beijing Jiaotong University**, School of Traffic and Transportation, Bachelor of Engineering, GPA: 3.80 09/2018 – 06/2022
Thesis: *Customized Bus Planning Method Responding to Spatiotemporal Concentrated Demand in Super-Peak.*

TECHNICAL SKILLS

- **Programming & Modeling Languages:** Python, MATLAB, C#, C++; GAMS.
- **Optimization Solvers & Methods:** mixed-integer programming, column generation, branch-and-price.
- **Machine Learning Frameworks:** TensorFlow, PyTorch.
- **Machine Learning Methods:** graph neural networks, sequence-to-sequence forecasting, transfer learning, multi-task learning.
- **Transportation & Infrastructure Modeling:** space-time network modeling, path-based traffic assignment, microscopic traffic simulation, pavement performance prediction, pavement maintenance optimization.
- **English:** TOEFL 112 (Reading 30, Listening 30, Speaking 25, Writing 27). 01/2025

SELECTED HONORS & PROFESSIONAL SERVICE

- Associate Member, PIARC Technical Committee 1.1 (World Road Association) 2026 – Present
- Second-Class Academic Excellence Scholarship, Tsinghua University 2024
- Outstanding Graduate of Beijing Municipality (Top 5%, 19/380) 2022
- Wu Fu-Zhenhua Transportation Education Excellent Student Award (Top 0.1%, 1/1520) 2021
- Zhan Tianyou Scholarship (Top 0.2%, 2/1520; Awarded Twice) 2020, 2021
- Zhixing Scholarship / President's Scholarship (Top 0.1%, 10/16800) 2020